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The Editors
Quantum Information Processing
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Re: Submission of the research article *Exact Diamond Balls, Antipodal Widths, and Query-Uniform Compression Bounds for Quantum Channels*

Dear Editors,

I am pleased to submit the manuscript cited above — a 46-page, single-author research article — for consideration in *Quantum Information Processing*.

The manuscript develops an exact, dimension-by-dimension theory of compression for quantum channels: for channels acting between fixed finite-dimensional systems, how many continuous latent parameters suffice to encode an arbitrary channel and reconstruct it with uniformly small diamond-norm error, when the encoder is continuous and the decoder unrestricted? This is a continuous analogue of nonlinear approximation theory, distinct from tomography, which counts channel uses, and from programmable processors, which fix a decoding device. The optimal error is characterized as the minimal worst-case radius of the encoder's fibers. Lower bounds follow from antipodal pairs and latent-collision topology via Borsuk–Ulam arguments and the deleted-product coindex — even input-independent output preparation forces error one below the output-state dimension — while the largest diamond-norm ball inside the channel body is determined exactly: centered at the completely depolarizing channel, it is at once the optimal center and the optimal constant decoder, with exact covering radius, so the two envelopes meet at the exact width at zero latent dimension. Constructive upper bounds from convex projection and balanced pinching bracket the intermediate regime and close the gap in the collapse case and from the affine dimension on, where exact reconstruction is possible. Every finite-query adaptive network induces a continuous final code, so the same bottlenecks hold uniformly in the number of queries, with a one-query informationally complete measurement attaining the injective threshold and an $O(N^{-1/2})$ repetition rate.

The question is a basic resource-theoretic one for quantum information: the memory cost of representing, simulating, and communicating noisy dynamics with guaranteed worst-case accuracy. The neighbouring literature is largely asymptotic or architecture-specific, whereas the manuscript identifies exact thresholds and certified two-sided bounds that hold for every continuous encoder of the stated latent type, with accuracy measured in the diamond norm, the standard single-shot distinguishability distance for channels. Such architecture-agnostic bounds delimit what any compression scheme can achieve.

A companion paper, *Exact Affine Geometry, Optimal Centres, Certified Compression Bounds, and Flag-Quotient Width Obstructions for Finite-Outcome Quantum Instruments*, is being submitted separately to the journal. It treats finite-outcome instruments and proves the flag-quotient flatness dichotomy invoked in the input-independent case of the present manuscript; the two papers are self-contained, cite each other explicitly, and can be reviewed independently.

The submission lies squarely within the journal's scope. *Quantum Information Processing* publishes experimental and theoretical research across the entire spectrum of quantum information science, and this is a pure quantum-information-theory contribution written in the field's standard formalism. The minimal latent dimension for faithful channel representation, and the topological obstructions to attaining it, are of direct interest to the journal's readership in channel simulation, quantum communication, and resource-efficient information processing.

I confirm that the manuscript is original, unpublished, and not under consideration elsewhere; I am its sole author, with no funding and no competing interests to declare, and the purely theoretical work generated no datasets.

Thank you for your consideration. I look forward to the outcome of the review.

Yours sincerely,

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